

Is the gender gap in AI closing? Evidence from Wikipedia

COMP3020 Social Web Analytics: Group Assignment

Group 35: Naomi, Amber, Tristan

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1 Introduction

Women remain under-represented in artificial intelligence (AI). This report uses Wikipedia to ask whether that gap is closing. We study 919 AI researchers with an English Wikipedia article and answer five research questions:

1. Has the share of women in AI increased across birth eras?
2. Are women recognised on Wikipedia later than men?
3. Has the language used to describe women and men changed over time?
4. Which AI subfields emerge from the biography text, and what share of each is women?
5. Are women central in the network of biographies that link to each other?

RQ1 and RQ2 use hypothesis tests, RQ3 uses text analysis, RQ4 uses clustering and RQ5 uses network analysis. All five use the same dataset and the same birth eras, so the results can be compared directly.

Code and data: <https://github.com/minhhien365/COMP-3020---Group-Assignment>

2 Analysis

2.1 Data collection

2.1.1 How the data was accessed

We collected all the data from three free Wikimedia APIs. None of them need an API key.

1. **Wikidata (SPARQL):** we used this to find the AI researchers, along with their gender, birth year and a short description.
2. **Wikipedia API:** we used this to get each person's article intro, article length, the date the article was created, and the links to other articles.
3. **Wikimedia page views API:** we used this to get how many times each article was viewed.

All three APIs return JSON, so we read them with `fromJSON()` from `jsonlite`. Wikimedia blocks programs that don't identify themselves or that send requests too fast, so we wrote a helper function that sends a User-Agent and retries a request up to three times. We also paused between requests.

The code below is not run when the report is knitted, because it takes about 25 minutes and needs the internet. We saved the results to `data/people.csv` and `data/edges.csv` and the report loads those instead.

```
library("jsonlite")

UA = "COMP3020-group-project (student project)" # identifies our program to Wikimedia

# download a URL, turn the JSON into R objects and retry up to 3 times
get_json = function(u) {
  for (attempt in 1:3) {
    result = tryCatch({
      con = url(u, headers = c("User-Agent" = UA))
      txt = paste(readLines(con, warn = FALSE, encoding = "UTF-8"), collapse = "")
      close(con)
      fromJSON(txt)
    }, error = function(e) NULL)
    if (!is.null(result)) return(result)
    Sys.sleep(2 * attempt) # wait longer before each retry
  }
  return(NULL)
}

WIKI = "https://en.wikipedia.org/w/api.php?format=json&formatversion=2"
```

2.1.2 Step 1: finding the AI researchers (Wikidata)

We kept every human with an English Wikipedia article whose field of work is artificial intelligence (or a sub-field of it, like machine learning), or whose occupation is AI researcher. The `/wdt:P279*` part of the query is what pulls in the sub-fields.

```
query = '
SELECT ?person ?article ?genderLabel ?description
      (MIN(YEAR(?dob)) AS ?birth_year) (SAMPLE(?sl) AS ?sitelinks)
WHERE {
  { ?person wdt:P101/wdt:P279* wd:Q11660 . }
  UNION
  { ?person wdt:P106/wdt:P279* wd:Q15976092 . }
  ?person wdt:P31 wd:Q5 ;
    wdt:P21 ?gender ;
    wikibase:sitelinks ?sl .
  OPTIONAL { ?person wdt:P569 ?dob . }
  OPTIONAL { ?person schema:description ?description .
    FILTER(LANG(?description) = "en") }
  ?article schema:about ?person ;
    schema:isPartOf <https://en.wikipedia.org/> .
  ?gender rdfs:label ?genderLabel . FILTER(LANG(?genderLabel) = "en")
}
GROUP BY ?person ?article ?genderLabel ?description
'

wd = get_json(paste0("https://query.wikidata.org/sparql?format=json&query=",
  URLEncode(query, reserved = TRUE)))
b = wd$results$bindings

people = data.frame(
  qid      = sub(".*/", "", b$person$value),
  title    = gsub("_", " ", URLdecode(sub(".*\\/wiki/", "", b$article$value))),
  gender   = b$genderLabel$value,
```

```

description = b$description$value,
birth_year = as.numeric(b$birth_year$value),
sitelinks = as.numeric(b$sitelinks$value),
stringsAsFactors = FALSE)
people = people[!duplicated(people$qid), ]

people$gender_group = ifelse(people$gender %in% c("female", "male"),
                             people$gender, "other/non-binary")

```

2.1.3 Step 2: article text, length and creation date (Wikipedia API)

We got the intro text and article length 20 articles at a time. For the creation date we asked for each article's first edit, which has to be done one article at a time. Recognition lag (for RQ2) is then the creation year minus the birth year.

```

# intro text and length, 20 articles per request
groups = split(1:nrow(people), ceiling(seq_len(nrow(people)) / 20))
for (g in groups) {
  u = paste0(WIKI, "&action=query&prop=extracts|info&exintro=1&explaintext=1&exlimit=20",
             "&titles=", URLencode(paste(people$title[g], collapse = "|"), reserved = TRUE))
  d = get_json(u)
  if (is.null(d)) next
  m = match(d$query$pages$title, people$title)
  people$intro_text[m[!is.na(m)]] = d$query$pages$extract[!is.na(m)]
  people$article_bytes[m[!is.na(m)]] = d$query$pages$length[!is.na(m)]
  Sys.sleep(0.5)
}

# creation date = timestamp of the first edit
for (i in 1:nrow(people)) {
  u = paste0(WIKI, "&action=query&prop=revisions&rvprop=timestamp&rvlimit=1&rvdir=newer",
             "&titles=", URLencode(people$title[i], reserved = TRUE))
  d = get_json(u)
  if (!is.null(d$query$pages$revisions[[1]]))
    people$created[i] = d$query$pages$revisions[[1]]$timestamp
  Sys.sleep(0.2)
}
people$created_year = as.numeric(substr(people$created, 1, 4))
people$recognition_lag = people$created_year - people$birth_year

```

2.1.4 Step 3: page views (Wikimedia page views API)

We added up the monthly views from October 2025 to September 2026. We only counted human readers, so bot traffic is left out.

```

for (i in 1:nrow(people)) {
  t = URLencode(gsub(" ", "_", people$title[i]), reserved = TRUE)
  u = paste0("https://wikimedia.org/api/rest_v1/metrics/pageviews/per-article/",
             "en.wikipedia/all-access/user/", t, "/monthly/20251001/20260930")
  d = get_json(u)
  people$pageviews_12m[i] = if (is.null(d)) 0 else sum(d$items$views)
  Sys.sleep(0.2)
}

```

2.1.5 Step 4: links between researchers (Wikipedia API)

For each person we got every article their page links to and only kept the links that go to someone else in our dataset. The API only gives 500 links per request, so for long articles we kept asking for the next batch until there were none left.

```
edges = data.frame(from = character(), to = character(), stringsAsFactors = FALSE)
for (i in 1:nrow(people)) {
  linked = c()
  cont = ""
  repeat {
    u = paste0(WIKI, "&action=query&generator=links&gplnamespace=0&gpllimit=max&redirects=1",
               "&titles=", URLencode(people$title[i], reserved = TRUE), cont)
    d = get_json(u)
    if (is.null(d)) break
    linked = c(linked, d$query$pages$title)
    if (is.null(d$continue$gplcontinue)) break
    cont = paste0("&gplcontinue=", URLencode(d$continue$gplcontinue, reserved = TRUE))
  }
  targets = setdiff(people$qid[people$title %in% linked], people$qid[i]) # no self-links
  if (length(targets) > 0) edges = rbind(edges, data.frame(from = people$qid[i], to = targets))
  Sys.sleep(0.2)
}
edges = unique(edges)

write.csv(people, "data/people.csv", row.names = FALSE, fileEncoding = "UTF-8")
write.csv(edges, "data/edges.csv", row.names = FALSE, fileEncoding = "UTF-8")
```

2.1.6 R packages

- jsonlite for reading the API data
- base R for the hypothesis tests and plots (RQ1, RQ2)
- tidyverse, tidytext and cluster for the text analysis and clustering (RQ3, RQ4)
- igraph for the network analysis (RQ5)

2.1.7 Variables and volume

```
table(people$gender_group)
```

```
##
##          female          male other/non-binary
##          218          699          2
```

```
nrow(edges)
```

```
## [1] 2204
```

The final dataset has 919 researchers (218 women, 699 men and 2 other/non-binary) and 2,204 links between their articles.

Variable	Description
qid, title	Wikidata ID and Wikipedia article title
gender_group	female, male or other/non-binary
birth_year, era	birth year and birth era (pre-1950, 1950-69, 1970-89, 1990+)
created_year	year the Wikipedia article was created
recognition_lag	created_year - birth_year (RQ2)
intro_text	the article's intro text (RQ3, RQ4)
article_bytes, pageviews_12m, sitelinks	article length, page views in the last year, number of language versions
in_degree, out_degree	links to / from other people in the dataset (RQ5)

2.1.8 Data cleaning

When Wikidata only knows someone was born in the 20th century, it saves their birth year as 2000. This happened for 21 people (for example Alex Krizhevsky and Ruslan Salakhutdinov), so we set their birth year to unknown. That leaves 201 people (22%) with no birth year. They are left out of anything split by era, which leaves 716 women and men.

2.2 Hypothesis testing (RQ1 and RQ2)

2.2.1 RQ1: Has the share of women in AI increased across birth eras?

Hypotheses.

- H0: Gender and birth era are independent.
- HA: Gender and birth era are not independent.

Test choice. Both variables are categorical, so we use a chi-squared test for independence on the era x gender table of counts. The built-in test needs every expected count to be greater than 5, so we check this first.

We first count the women and men in each birth era, convert these counts to percentages, and check the expected counts.

```
# writes p-values as "< 0.001" or "= 0.94" in the text
fmt_p = function(p) if (p < 0.001) "< 0.001" else paste("=", signif(p, 2))

tab = table(known$era, known$gender_group)
tab
```

```
##
##           female male
## pre-1950      23  154
## 1950-69       60  264
## 1970-89       56  135
## 1990+         2   22
```

```
# share of women and men in each era (%)
round(prop.table(tab, 1) * 100, 1)
```

```
##
##           female male
## pre-1950    13.0 87.0
## 1950-69     18.5 81.5
## 1970-89     29.3 70.7
## 1990+       8.3 91.7
```

```
# assumption check: expected counts
ct = chisq.test(tab)
round(ct$expected, 1)
```

```
##
##           female male
## pre-1950    34.9 142.1
## 1950-69     63.8 260.2
## 1970-89     37.6 153.4
## 1990+       4.7  19.3
```

Women are a minority in every era. All expected counts are above 5 except women born 1990+ (4.7), because that era has only 24 people. So we need simulation.

```
ct = chisq.test(tab, simulate.p.value = TRUE)
ct
```

```
##
## Pearson's Chi-squared test with simulated p-value (based on 2000
## replicates)
```

```
##
## data: tab
## X-squared = 18.455, df = NA, p-value = 0.0009995
```

The p-value is < 0.001 , so we reject H_0 : the share of women differs between eras. The test does not show how it differs, so we plot the share of women in each era below. The error bars are bootstrap 95% confidence intervals. Wider bars mean less certainty, which happens when an era has fewer people.

```
# share of women in each era (%) and number of people in each era
share = prop.table(tab, 1)[, "female"] * 100
n = rowSums(tab)

# bootstrap 95% confidence interval for each era
lower = c()
upper = c()
for (e in levels(known$era)) {
  women = known$gender_group[known$era == e] == "female" # TRUE/FALSE for each person
  boot = replicate(1000, mean(sample(women, replace = TRUE))) * 100
  lower = c(lower, quantile(boot, 0.025))
  upper = c(upper, quantile(boot, 0.975))
}

# line chart with error bars
plot(1:4, share, type = "b", pch = 19, col = "darkorange", ylim = c(0, 40), xaxt = "n",
     main = "Share of women in AI by birth era", xlab = "Birth era", ylab = "Women (%)")
axis(1, at = 1:4, labels = paste0(names(share), " (n=", n, ")"))
arrows(1:4, lower, 1:4, upper, angle = 90, code = 3, length = 0.05)
```

Share of women in AI by birth era

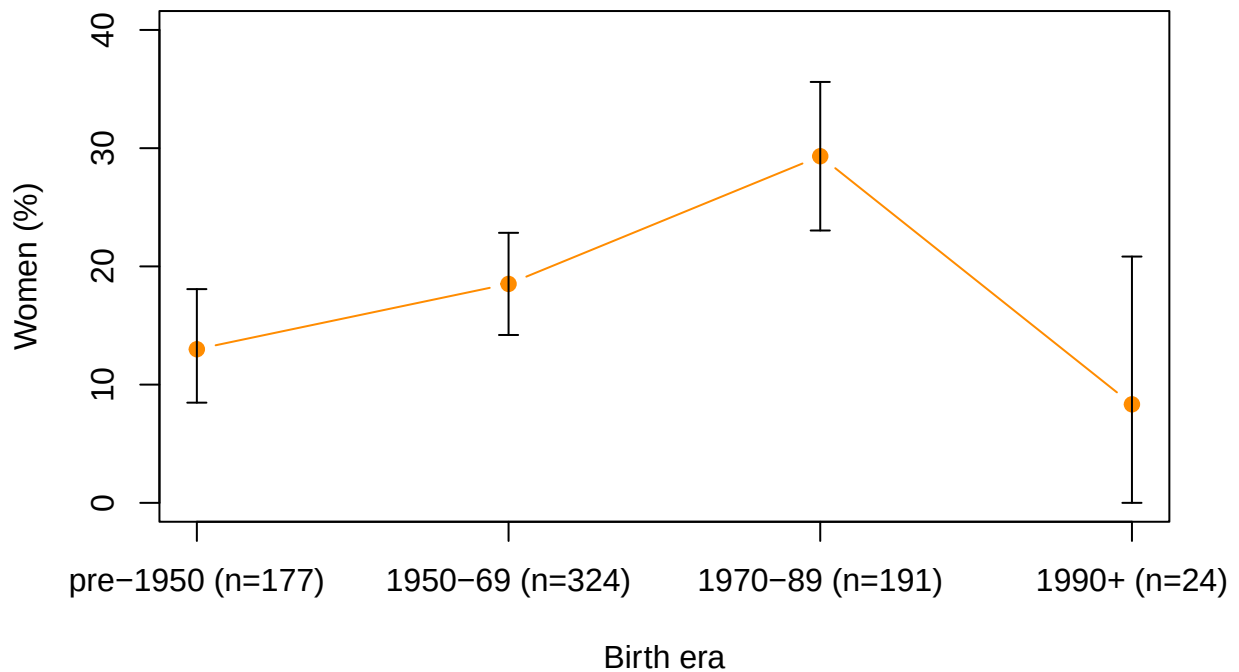


Figure 1: Share of women among AI researchers by birth era. Error bars are bootstrap 95% confidence intervals.

```
# save a copy for the poster
dev.copy(png, file.path(fig_dir, "rq1_share_women_by_era.png"), width = 1400, height = 900, res = 200)

## png
## 3

invisible(dev.off())
```

The share of women rises from 13% (pre-1950) to 19% (1950-69) and 29% (1970-89), and the error bars for pre-1950 (8-18%) and 1970-89 (23-36%) do not overlap. The drop to 8% for 1990+ rests on only 24 people (2 women), so its error bar is very wide (0-21%) and it should not be read as a reversal. Younger researchers have also had less time to become notable.

2.2.2 RQ2: Are women recognised on Wikipedia later than men?

To measure how quickly a person is recognised on Wikipedia, we use recognition lag = year the article was created - birth year. For example, someone born in 1960 whose article was created in 2010 has a recognition lag of 50 years. A shorter lag means the person was recognised earlier in their life. If women in AI are overlooked, we would expect them to receive an article later in life than men, so their recognition lag would be longer.

Recognition lag has two important limitations. First, Wikipedia was only launched in 2001, so anyone born long before then automatically has a big lag, whatever their gender. This means lag depends strongly on birth era, so we also compare women and men within each era. Second, the creation date reflects when a Wikipedia editor chose to write the article, not when the person achieved something notable. This is exactly why it is useful here: it measures when Wikipedia recognised them, not their career itself.

Hypotheses.

- H0: There is no difference in mean recognition lag between women and men.
- HA: There is a difference in mean recognition lag between women and men.

Test choice. We use a randomisation test for the difference in means and a Wilcoxon rank-sum test as a check.

We first separate the recognition lags of women and men and compare their size, mean and median.

```
lagF = known$recognition_lag[known$gender_group == "female"]
lagM = known$recognition_lag[known$gender_group == "male"]

data.frame(group = c("Women", "Men"),
           n      = c(length(lagF), length(lagM)),
           mean   = round(c(mean(lagF), mean(lagM)), 1),
           median = c(median(lagF), median(lagM)))

##   group    n mean median
## 1 Women 141 53.6     51
## 2   Men 575 53.5     52
```

Women and men have almost identical recognition lags overall: a mean of 53.6 vs 53.5 years, and a median of 51 vs 52 years. There are far more men (575) than women (141) with a known birth year.

To test whether this difference could have occurred by chance, we run the randomisation test: if H0 is true, the gender labels do not matter, so we shuffle them 2000 times and recalculate the difference in means each time.

```
obs_diff = mean(lagF) - mean(lagM) # women minus men
combined = c(lagF, lagM)
nF = length(lagF)

rand_diff = replicate(2000, {
  shuffled = sample(combined)
  mean(shuffled[1:nF]) - mean(shuffled[-(1:nF)])
})

# two-sided p-value: proportion of random differences at least as extreme as observed
```

```
p_rand = mean(abs(rand_diff) >= abs(obs_diff))
c(observed_difference = round(obs_diff, 2), p_value = p_rand)
```

```
## observed_difference      p_value
##              0.110          0.943
hist(rand_diff, breaks = 40, col = "grey80", main = "",
      xlab = "Difference in sample means (women - men, years)")
abline(v = obs_diff, col = "red", lwd = 2)
```

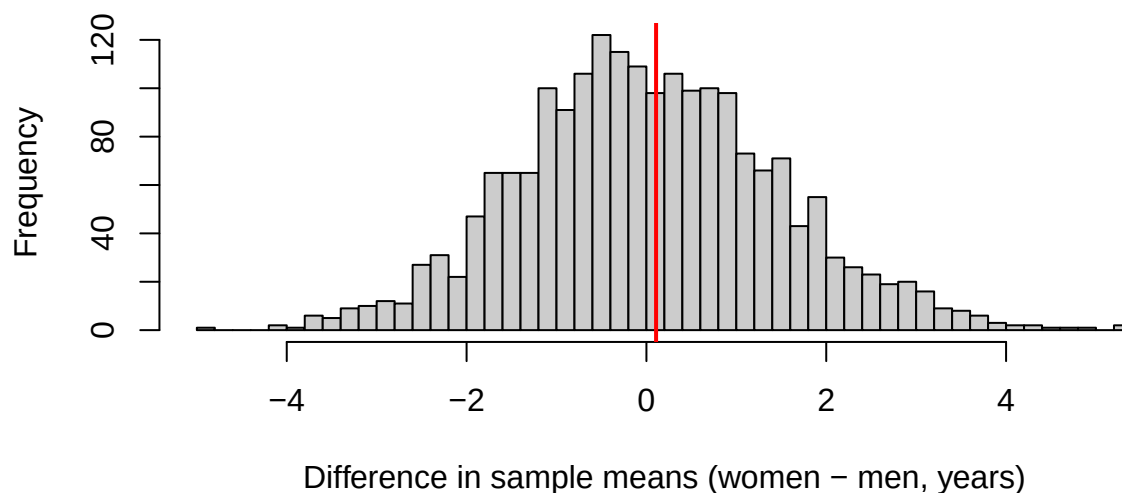


Figure 2: Randomisation distribution of the difference in mean recognition lag (women minus men) under H_0 . The red line is the observed difference.

The observed difference in means is only 0.11 years. In the histogram, the red line sits in the middle of the randomisation distribution, so a difference this small easily happens by chance. The p-value is 0.943, so we do not reject H_0 : overall there is no evidence that women's lag differs from men's.

Recognition lag is skewed, with a few people having very long lags that can pull the mean up or down. As a check, we also run a Wilcoxon rank-sum test, which compares the ranks of the lags instead of their actual values.

```
wt = wilcox.test(lagF, lagM)
wt

##
## Wilcoxon rank sum test with continuity correction
##
## data: lagF and lagM
## W = 39886, p-value = 0.7672
## alternative hypothesis: true location shift is not equal to 0
```

The Wilcoxon test agrees: $W = 39885.5$, $p = 0.77$, so there is no overall difference.

Because lag depends strongly on birth era, we repeat the Wilcoxon test separately within each era to compare like with like.

```
era_tab = data.frame(era = levels(known$era), median_women = NA, median_men = NA, p_value = NA)

for (i in 1:4) {
```



```

f = known$recognition_lag[known$era == era_tab$era[i] & known$gender_group == "female"]
m = known$recognition_lag[known$era == era_tab$era[i] & known$gender_group == "male"]
era_tab$median_women[i] = median(f)
era_tab$median_men[i]   = median(m)
era_tab$p_value[i]      = wilcox.test(f, m, exact = FALSE)$p.value
}

era_show = era_tab
era_show$p_value = format.pval(era_tab$p_value, digits = 2, eps = 0.001)
knitr::kable(era_show, digits = 1,
              caption = "Median recognition lag (years) by gender within each birth era, with Wilcoxon rank-sum p-values")

```

Table 2: Median recognition lag (years) by gender within each birth era, with Wilcoxon rank-sum p-values.

era	median_women	median_men	p_value
pre-1950	75.0	70.0	0.011
1950-69	57.0	51.5	<0.001
1970-89	40.0	40.0	0.931
1990+	27.5	31.0	0.793

Splitting by era changes the picture. Women born before 1950 waited longer than men (median 75 vs 70 years, $p = 0.011$), and so did women born 1950-69 (57 vs 51.5 years, $p < 0.001$). For 1970-89 (40 vs 40, $p = 0.93$) and 1990+ (27.5 vs 31, $p = 0.79$) there is no significant difference.

Finally, we plot the lags of women and men side by side within each era, to see both the effect of era and any gender gap within it.

```

boxplot(recognition_lag ~ gender_group + era, data = known,
        col = c("pink", "lightblue"), at = c(1, 2, 4, 5, 7, 8, 10, 11), xaxt = "n",
        main = "Recognition lag by gender and birth era",
        xlab = "Birth era", ylab = "Years from birth to Wikipedia article")
axis(1, at = c(1.5, 4.5, 7.5, 10.5), labels = levels(known$era))
legend("topright", legend = c("Female", "Male"), fill = c("pink", "lightblue"))

# save a copy for the poster
dev.copy(png, file.path(fig_dir, "rq2_lag_by_gender_era.png"), width = 1600, height = 900, res = 200)

## png
## 3

invisible(dev.off())

```

The boxplot shows lag falling sharply from older to younger eras for both genders, mostly because Wikipedia only began in 2001. In the two oldest eras the women's boxes sit higher than the men's, while from 1970 onwards they overlap almost completely. The overall test hides the older gap because women are concentrated in the younger eras, which have shorter lags.

2.3 Text analysis (RQ3)

RQ3 asks whether the language used to describe women and men in AI differs, and whether it has changed across birth eras. We use the intro of each Wikipedia article, because it is the part every article has and it summarises why the person is notable.

```

library(tidyverse) # data wrangling and ggplot2 charts
library(tidytext)  # tokenising, stop words and tf-idf
library(cluster)   # silhouette widths (RQ4)

# women and men only, with the same birth eras as RQ1 and RQ2

```

Recognition lag by gender and birth era

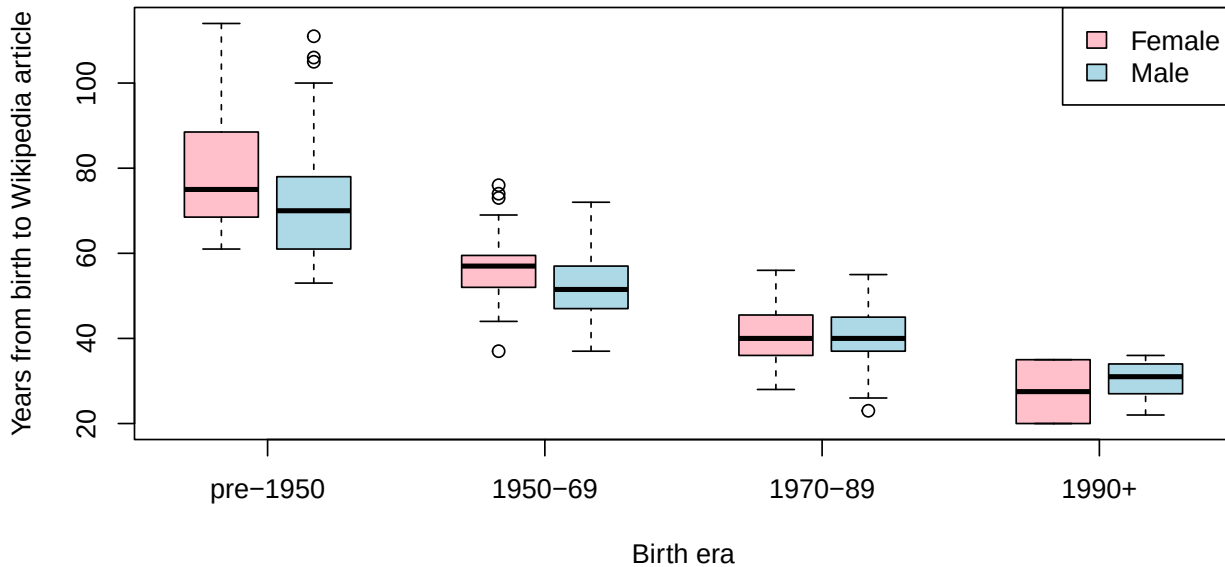


Figure 3: Recognition lag (years from birth to Wikipedia article) by gender and birth era.

```

bios = as_tibble(people) |>
  rename(wikidata_id = qid, name = title, intro = intro_text) |>
  filter(gender %in% c("female", "male")) |>
  mutate(era = factor(era, levels = c("pre-1950", "1950-69", "1970-89", "1990+")),
         gender = factor(gender, levels = c("female", "male")))

```

2.3.1 Preparing the text

We remove people with an empty intro, then split each intro into single words. We drop numbers, words of one or two letters, common English stop words, three uninformative words (born, also, known) and the words of each person's own name, so that names do not dominate the results.

```

people_text = bios |> filter(!is.na(intro), str_trim(intro) != "")

# words of each person's own name, removed from their own intro
name_tokens = people_text |>
  select(wikidata_id, name) |>
  unnest_tokens(word, name) |>
  distinct()

custom_stop = tibble(word = c("born", "also", "known"))

tokens = people_text |>
  select(wikidata_id, gender, era, intro) |>
  unnest_tokens(word, intro) |>
  filter(!str_detect(word, "[0-9]"), str_length(word) > 2) |>
  anti_join(stop_words, by = "word") |>
  anti_join(custom_stop, by = "word") |>
  anti_join(name_tokens, by = c("wikidata_id", "word"))

```

Each intro is one document, represented as a bag of words with tf-idf weights. We keep words used at least 5

times, so that rare words and typos do not add noise. This matrix is also the input for the clustering in RQ4.

```
word_counts = tokens |> count(wikidata_id, word)
tfidf_long = word_counts |> bind_tf_idf(word, wikidata_id, n)
keep_words = word_counts |> count(word) |> filter(n >= 5) |> pull(word)

tfidf_mat = tfidf_long |>
  filter(word %in% keep_words) |>
  cast_sparse(wikidata_id, word, tf_idf) |>
  as.matrix()

tibble(people_with_text = n_distinct(tokens$wikidata_id),
       total_words      = nrow(tokens),
       distinct_words   = n_distinct(tokens$word),
       words_kept       = length(keep_words))
```

```
## # A tibble: 1 x 4
##   people_with_text total_words distinct_words words_kept
##   <int>          <int>          <int>          <int>
## 1           909      40265          7556          1127
```

length of the cleaned intros, women vs men

```
tokens |>
  count(wikidata_id, gender) |>
  group_by(gender) |>
  summarise(people = n(), median_words = median(n), .groups = "drop")
```

```
## # A tibble: 2 x 3
##   gender people median_words
##   <fct>   <int>         <dbl>
## 1 female   218           33
## 2 male    691           35
```

7 people have no usable intro, which leaves 909 people. Women's and men's intros have a similar length after cleaning, so differences in word use are not caused by length.

2.3.2 How are women and men described?

```
plot_freq = tokens |>
  count(word, sort = TRUE) |>
  slice_head(n = 20) |>
  ggplot(aes(n, reorder(word, n))) +
  geom_col() +
  labs(title = "Most frequent words in AI biographies",
       x = "Number of occurrences", y = NULL)
plot_freq

ggsave("figures/freq_words_overall.png", plot_freq, width = 6, height = 5, dpi = 300)
```

The most frequent words are computer, university, professor, intelligence and research, so people in AI are described mainly by their academic positions and affiliations. American is also in the top ten.

There are far more men than women, so the next chart uses each word's share of all words within a gender instead of raw counts.

```
plot_gender = tokens |>
  count(gender, word) |>
  group_by(gender) |>
  mutate(share = n / sum(n)) |>
  slice_max(share, n = 12, with_ties = FALSE) |>
  ungroup() |>
  ggplot(aes(share, reorder_within(word, share, gender), fill = gender)) +
```

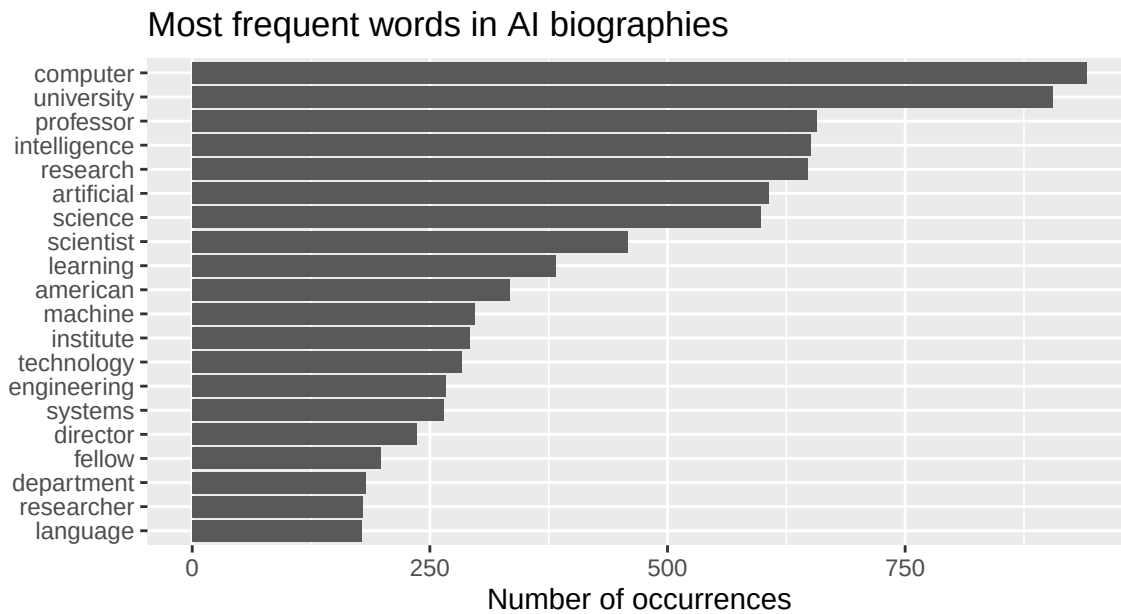


Figure 4: The 20 most frequent words in AI biographies.

```
geom_col(show.legend = FALSE) +
scale_y_reordered() +
facet_wrap(~gender, scales = "free_y") +
labs(title = "Most frequent words: women vs men",
x = "Share of all words", y = NULL)
plot_gender
```

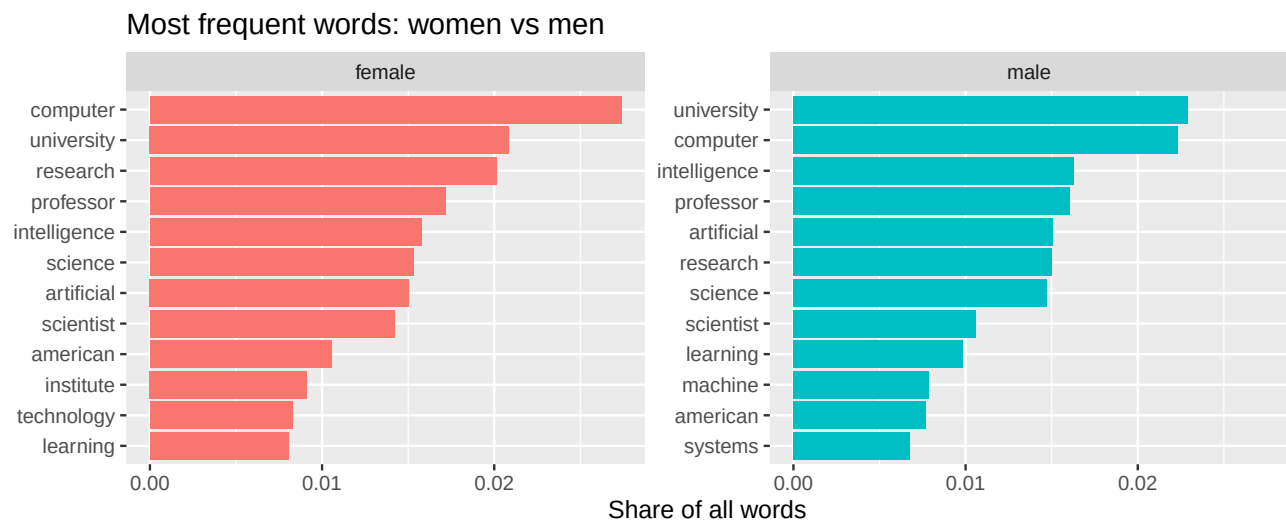


Figure 5: The 12 most frequent words for women and for men, as a share of all words within each gender.

```
ggsave("figures/freq_words_gender.png", plot_gender, width = 8, height = 5, dpi = 300)
```

Women and men share a very similar vocabulary. University, research, professor and intelligence are in both top twelves. The differences are small: research, scientist, institute and technology take a slightly larger share of women's intros, while machine and systems appear among men's top words but not women's.

To see how the language changes over time, we treat each era and gender group as one document and calculate tf-idf across the groups. This picks out the words that are typical of one group and rare in the others. A word only counts if at least 3 different people in the group use it, so one long biography cannot dominate a group. Women born 1990 or later are left out because there are only 2 of them.

```
min_people = 3

group_sizes = tokens |>
  filter(!is.na(era)) |>
  distinct(wikidata_id, era, gender) |>
  count(era, gender, name = "group_n") |>
  filter(group_n >= 10) # drops women born 1990+ (2 people)

plot_era = tokens |>
  filter(!is.na(era)) |>
  semi_join(group_sizes, by = c("era", "gender")) |>
  mutate(group = paste(era, gender, sep = " | ")) |>
  group_by(group, word) |>
  summarise(n = n(), n_people = n_distinct(wikidata_id), .groups = "drop") |>
  bind_tf_idf(word, group, n) |>
  filter(n_people >= min_people, tf_idf > 0) |>
  group_by(group) |>
  slice_max(tf_idf, n = 6, with_ties = FALSE) |>
  ungroup() |>
  left_join(group_sizes |>
    mutate(group = paste(era, gender, sep = " | "),
           group_label = paste0(group, " (n=", group_n, ")")) |>
    select(group, group_label),
    by = "group") |>
  ggplot(aes(tf_idf, reorder_within(word, tf_idf, group_label))) +
  geom_col() +
  scale_y_reordered() +
  facet_wrap(~group_label, scales = "free_y", ncol = 2) +
  labs(title = "Most distinctive words by era and gender (tf-idf)",
       x = "tf-idf", y = NULL)
plot_era

ggsave("figures/distinctive_words_era_gender.png", plot_era, width = 8, height = 10, dpi = 300)
```

The distinctive words change across eras. Women born before 1950 are the only group whose top word is women, because several of their intros describe them as advocates for women in science. Women born 1950-69 are linked to research topics (affective, bayesian, semantic web), while women born 1970-89 are linked to policy and leadership positions (alliance, roles, influential, lead). Men born before 1950 are described with classic AI terms (fuzzy, recognition, lisp), and men born 1990 or later with industry and AI safety terms (openai, anthropic, risks, company, models). Science appears in several groups only because one group rarely uses it. These comparisons are descriptive and were not tested.

2.4 Clustering (RQ4)

RQ4 asks which AI subfields emerge from the biography text and what share of each is women. We cluster people by the words in their intro, so the subfields come from the data rather than from a list we chose.

2.4.1 Representation and number of clusters

Each person is a row of tf-idf weights over the kept words. Each row is scaled to length 1 so that longer intros do not look different just because of their length. With unit-length rows, Euclidean distance ranks pairs of people the same way as cosine distance, so k-means groups people with similar word use.

```
row_norm = sqrt(rowSums(tfidf_mat^2))
m_norm   = tfidf_mat[row_norm > 0, , drop = FALSE] / row_norm[row_norm > 0]
```

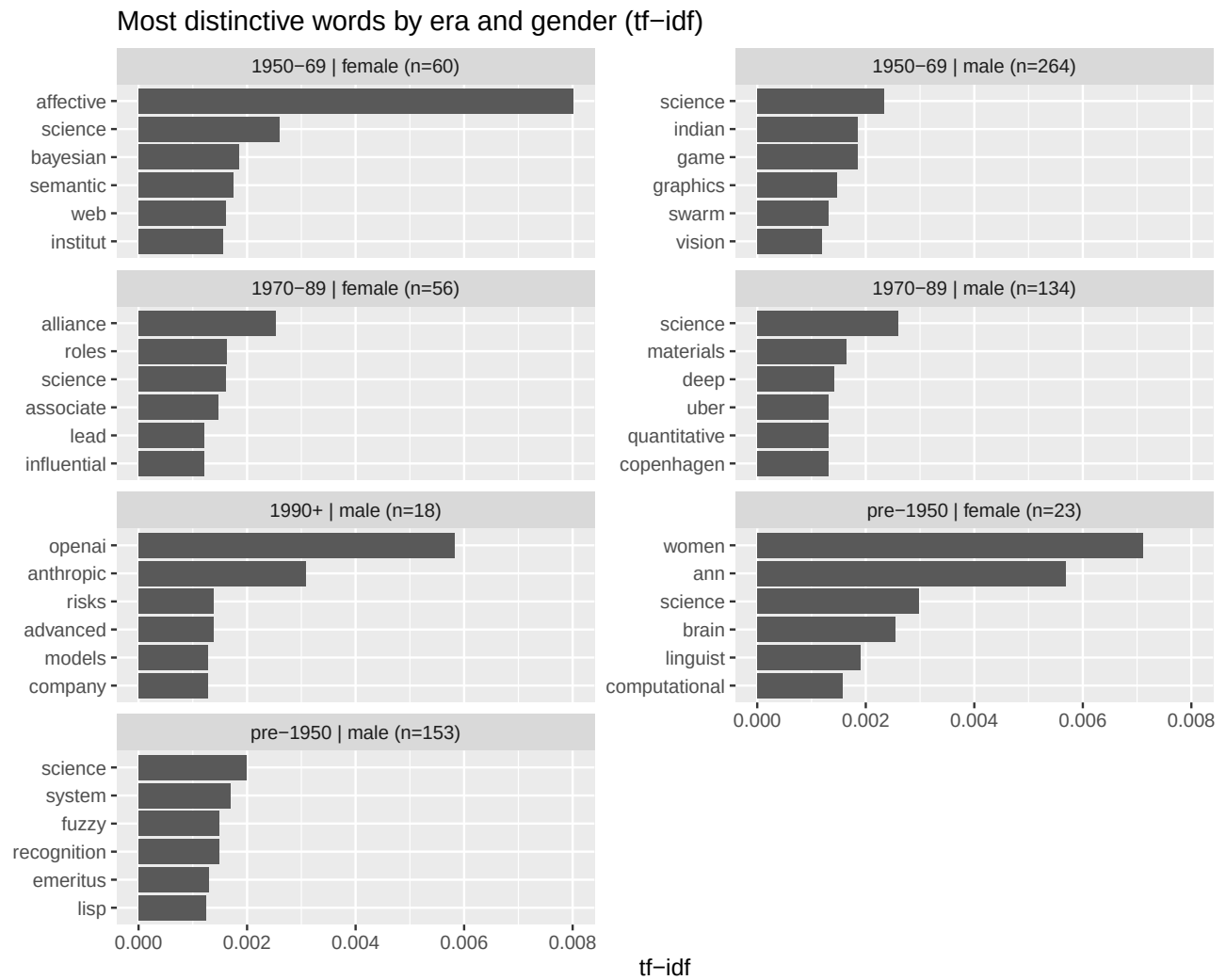


Figure 6: The most distinctive words (tf-idf) for each birth era and gender group with at least 10 people. Only words used by at least 3 people in the group are shown.

```
dim(m_norm)
```

```
## [1] 909 1127
```

We run k-means for $k = 2$ to 10 and compare the within-cluster sum of squares (elbow plot) and the average silhouette width.

```
set.seed(1)
ks = 2:10
fits = map(ks, \(k) kmeans(m_norm, centers = k, nstart = 10, iter.max = 50))
d = dist(m_norm)

k_check = tibble(k = ks,
  elbow = map_dbl(fits, "tot.withinss"),
  silhouette = map_dbl(fits, \(f) mean(silhouette(f$cluster, d)[, 3])))

p_k = k_check |>
  pivot_longer(c(elbow, silhouette), names_to = "measure") |>
  ggplot(aes(k, value)) +
  geom_line() + geom_point() +
  facet_wrap(~measure, scales = "free_y") +
  scale_x_continuous(breaks = ks) +
  labs(title = "Choosing the number of clusters", y = NULL)
p_k
```

Choosing the number of clusters

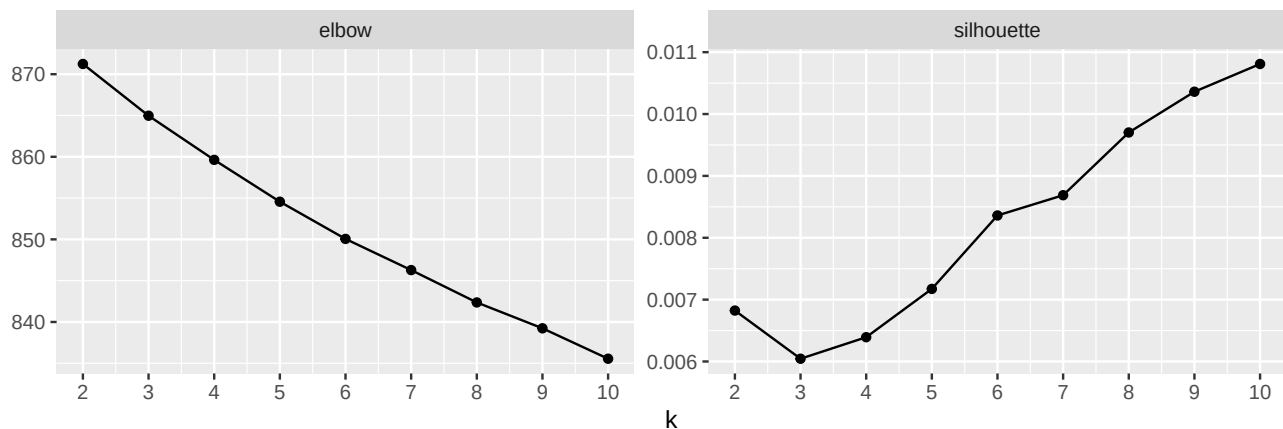


Figure 7: Elbow plot and average silhouette width for $k = 2$ to 10.

```
ggsave("figures/choose_k.png", p_k, width = 8, height = 4, dpi = 300)
```

The elbow plot has no clear bend and the silhouette width is very low for every k (at most 0.011), so the clusters overlap and no k is clearly best. We choose $k = 5$ because the clusters are easy to interpret and large enough to compare by gender and era.

```
K = 5
set.seed(1)
km = kmeans(m_norm, centers = K, nstart = 25, iter.max = 100)

# top words of each cluster centre and the 3 people closest to it
ctr_norm = km$centers / sqrt(rowSums(km$centers^2))
cos_sims = m_norm %*% t(ctr_norm)

cluster_words = map_dfr(seq_len(K), \(i) {
  members = which(km$cluster == i)
```

```

closest = members[order(cos_sims[members, i], decreasing = TRUE)][1:3]
tibble(cluster = i,
       size = length(members),
       top_words = paste(names(sort(km$centers[i, ], decreasing = TRUE))[1:6], collapse = ", "),
       closest = paste(bios$name[match(rownames(m_norm)[closest], bios$wikidata_id)],
                      collapse = ", "))
})
knitr::kable(cluster_words, caption = "Size, top words and the 3 most typical people of each cluster.")

```

Table 3: Size, top words and the 3 most typical people of each cluster.

cluster	size	top_words	closest
1	143	learning, machine, statistics, neural, computer, science	Barbara Hammer (computer scientist), Frank Hutter, John Shawe-Taylor
2	68	language, linguistics, natural, computational, processing, linguist	Vera Demberg, Emily M. Bender, Lori Levin
3	489	artificial, computer, science, intelligence, university, research	Nello Cristianini, Takayuki Ito, Rolf Pfeifer
4	88	electrical, engineering, ieee, fellow, electronics, engineers	Hao Ying, Gabor Fichtinger, Fakhreddine Karray
5	121	company, founder, chinese, founded, entrepreneur, ceo	Robin Li, Yan Junjie, Amnon Shashua

```

# name each cluster from its top 10 words (k-means can number the clusters differently
# on another computer, so the names are not tied to the cluster numbers)
cluster_names = sapply(seq_len(K), \(i) {
  w = names(sort(km$centers[i, ], decreasing = TRUE))[1:10]
  if (any(c("language", "linguistics") %in% w[1:3])) "Language & linguistics"
  else if (any(c("founder", "company", "entrepreneur") %in% w)) "Founders & entrepreneurs"
  else if ("electrical" %in% w) "Electrical engineering"
  else if (any(c("learning", "statistics") %in% w[1:3])) "Machine learning & statistics"
  else "General AI & computer science"
})
cluster_names = make.unique(cluster_names, sep = " ")
sizes_k = table(cluster_names[km$cluster]) # cluster sizes, used in the text
cluster_names

```

```

## [1] "Machine learning & statistics" "Language & linguistics"
## [3] "General AI & computer science" "Electrical engineering"
## [5] "Founders & entrepreneurs"

```

Four clusters are clear subfields or roles: machine learning and statistics ($n = 143$), language and linguistics ($n = 68$), electrical engineering ($n = 88$) and founders and entrepreneurs ($n = 121$), which groups people by a business role rather than a research topic. The largest cluster, general AI and computer science ($n = 489$), is a catch-all for academics described in general AI terms.

```

clusters = tibble(wikidata_id = rownames(m_norm), cluster = km$cluster) |>
  left_join(select(bios, wikidata_id, name, gender, era), by = "wikidata_id") |>
  mutate(cluster_label = factor(cluster, levels = seq_len(K), labels = cluster_names))

pc = prcomp(m_norm, rank. = 2)
p_clusters = clusters |>
  mutate(PC1 = pc$x[, 1], PC2 = pc$x[, 2]) |>
  ggplot(aes(PC1, PC2, colour = cluster_label)) +
  geom_point(alpha = 0.6) +
  labs(title = "AI biographies clustered by intro text (PCA view)", colour = "Cluster")
p_clusters

```

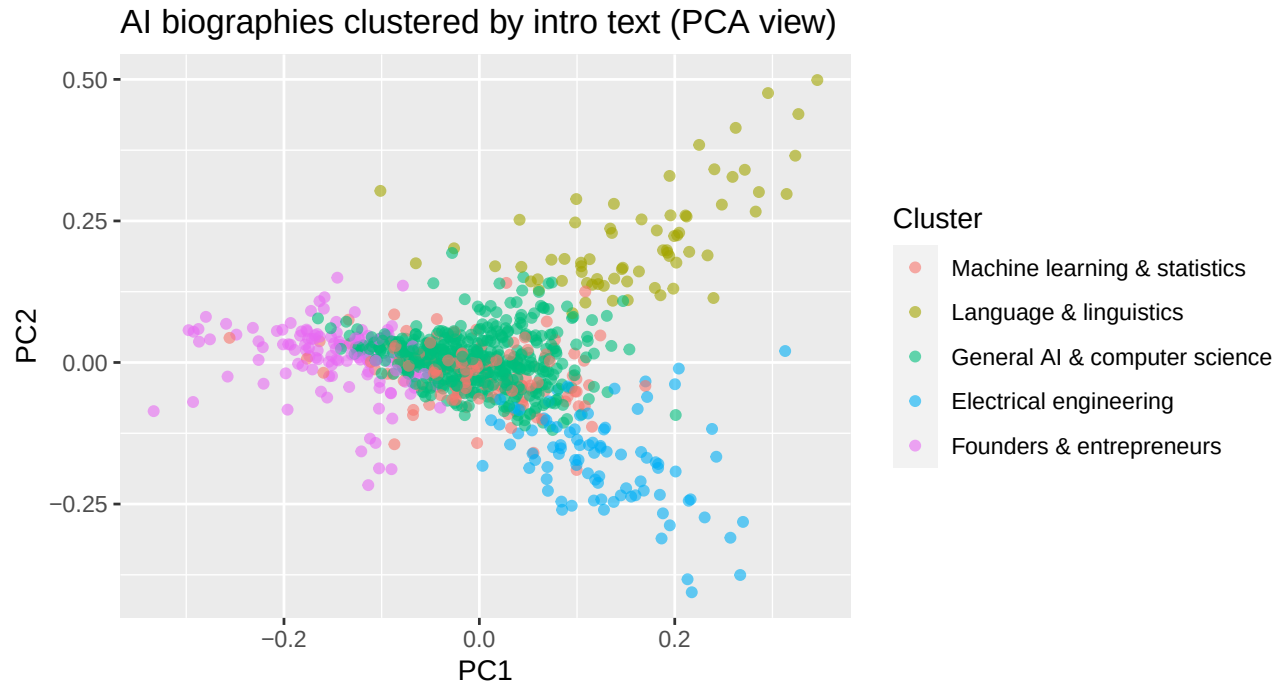



Figure 8: The clusters drawn on the first two principal components of the tf-idf matrix.

```
ggsave("figures/clusters_pca.png", p_clusters, width = 7, height = 5, dpi = 300)
saveRDS(clusters, "data/clusters.rds")
```

PCA reduces the many word dimensions to two so the clusters can be drawn. It only shows part of the structure, but the heavy overlap in the centre matches the low silhouette widths.

2.4.2 Women and men across clusters

H0: cluster membership is independent of gender.

HA: the two are not independent.

```
tab = table(clusters$cluster_label, clusters$gender)
tab
```

```
##
##               female male
## Machine learning & statistics    25  118
## Language & linguistics           31   37
## General AI & computer science    122  367
## Electrical engineering           16   72
## Founders & entrepreneurs         24   97
```

```
share_w = round(100 * prop.table(tab, 1)[, "female"]) # share of women in each cluster (%)
share_w
```

```
## Machine learning & statistics    Language & linguistics
##               17                  46
## General AI & computer science    Electrical engineering
##               25                  18
## Founders & entrepreneurs
##               20
```

```
ct4 = chisq.test(tab)
min(ct4$expected) # all expected counts must be at least 5
```

```
## [1] 16.30803
```

```
ct4
```

```
##
## Pearson's Chi-squared test
##
## data:  tab
## X-squared = 23.742, df = 4, p-value = 8.996e-05
```

Women are 24% of the people with text, but 46% of language and linguistics and only 17% of machine learning and statistics. All expected counts are above 5, and the association is significant ($\chi^2 = 23.7$, $df = 4$, $p < 0.001$), so we reject H_0 . Women are spread unevenly across the subfields.

Finally we look at women's share in each cluster by birth era. Groups with fewer than 10 people are left out because their shares are unreliable.

```
share_cluster_era = clusters |>
  filter(!is.na(era)) |>
  group_by(cluster_label, era) |>
  summarise(n = n(), share_women = mean(gender == "female"), .groups = "drop") |>
  filter(n >= 10)

# share of women (%) by cluster and era, with group sizes
share_cluster_era |>
  mutate(cell = paste0(round(100 * share_women), "% (n=", n, ")")) |>
  select(cluster_label, era, cell) |>
  pivot_wider(names_from = era, values_from = cell, values_fill = "-") |>
  knitr::kable(caption = "Share of women by cluster and birth era (groups of 10 or more).")
```

Table 4: Share of women by cluster and birth era (groups of 10 or more).

cluster_label	pre-1950	1950-69	1970-89
Machine learning & statistics	0% (n=13)	16% (n=43)	19% (n=43)
Language & linguistics	38% (n=16)	16% (n=19)	-
General AI & computer science	12% (n=116)	24% (n=189)	27% (n=85)
Electrical engineering	13% (n=23)	9% (n=33)	36% (n=14)
Founders & entrepreneurs	-	2% (n=40)	34% (n=41)

```
p_share = ggplot(share_cluster_era, aes(era, share_women)) +
  geom_col() +
  geom_text(aes(label = paste0("n=", n)), vjust = -0.3, size = 3) +
  scale_y_continuous(labels = scales::percent, limits = c(0, 1.1),
    breaks = c(0, 0.25, 0.5, 0.75, 1)) +
  facet_wrap(~cluster_label) +
  labs(title = "Share of women in each cluster, by birth era",
    x = "Birth era", y = "Share of women") +
  theme(axis.text.x = element_text(angle = 45, hjust = 1))
# chart saved for the poster only, the table above shows the same numbers
ggsave("figures/share_women_cluster_era.png", p_share, width = 8, height = 6, dpi = 300)
```

Women's share is higher for people born 1970-89 than for those born before 1950 in general AI and computer science (12% to 27%) and electrical engineering (13% to 36%, from a small group), and it rises from 2% to 34% among founders and entrepreneurs born 1950-69 and 1970-89. Machine learning and statistics stays below 20% in every era, and language and linguistics has too few recent researchers to show a trend. These patterns are descriptive and based on small groups.

2.5 Network analysis (RQ5)

RQ5 asks whether women are central in the network of AI biographies, or whether they sit at its edges.

2.5.1 The link network

Nodes are the 919 AI researchers. There is an edge from A to B when A's English Wikipedia article links to B's article, keeping only links to other people in the dataset. The graph is directed, because a link on A's page does not mean B's page links back, and PageRank needs that direction.

```
library(igraph)

# every edge points to someone in the dataset, with no self-links or duplicates
c(in_dataset = all(c(edges$from, edges$to) %in% people$qid),
  self_links = sum(edges$from == edges$to),
  duplicates = sum(duplicated(edges)))

## in_dataset self_links duplicates
##          1          0          0

g = graph_from_data_frame(edges, directed = TRUE,
  vertices = people[, c("qid", "title", "gender_group", "era")])
summary(g)

## IGRAPH a1851da DN-- 919 2204 --
## + attr: name (v/c), title (v/c), gender_group (v/c), era (v/c)
# share of links that go both ways (A -> B and B -> A)
mean(paste(edges$to, edges$from) %in% paste(edges$from, edges$to))

## [1] 0.784029
```

The network has 919 nodes and 2,204 directed edges. 78% of the links go both ways, mostly because of Wikipedia navigation boxes (navboxes), which link everyone on a list to everyone else.

2.5.2 Network properties

```
comp_w = components(g, mode = "weak")
lcc = induced_subgraph(g, which(comp_w$membership == which.max(comp_w$csizes)))

data.frame(density      = round(edge_density(g), 4),
  isolates             = sum(degree(g) == 0),
  weak_components      = comp_w$no,
  largest_comp         = vcount(lcc),
  lcc_edges            = ecount(lcc),
  lcc_diameter         = diameter(lcc, directed = FALSE),
  lcc_mean_dist        = round(mean_distance(lcc, directed = FALSE), 2))

## density isolates weak_components largest_comp lcc_edges lcc_diameter
## 1 0.0026 474 508 365 2134 14
## lcc_mean_dist
## 1 4.3
```

The network is very sparse (density 0.0026). 474 researchers (52%) have no links at all. Most of the rest form one large weakly connected component of 365 people that holds 97% of the links, with a diameter of 14 and an average distance of 4.3 links. Centrality is calculated on the full graph, and the largest component is only used for distances and the network plot.

```
iso_tab = table(V(g)$gender_group, ifelse(degree(g) > 0, "connected", "isolated"))
round(prop.table(iso_tab, 1), 2)

##
## connected isolated
```

```
## female          0.41    0.59
## male           0.51    0.49
## other/non-binary 0.00    1.00
```

```
ct_iso = chisq.test(iso_tab[c("female", "male"), ])
min(ct_iso$expected)
```

```
## [1] 105.7906
```

```
ct_iso
```

```
##
## Pearson's Chi-squared test with Yates' continuity correction
##
## data: iso_tab[c("female", "male"), ]
## X-squared = 5.6328, df = 1, p-value = 0.01763
```

H0: whether a researcher is connected is independent of gender.

HA: the two are not independent.

59% of women are isolated compared with 49% of men ($\chi^2 = 5.63$, $df = 1$, $p = 0.018$), so we reject H0. The article age check at the end of this section shows that this gap comes from women's articles being newer.

```
dd = degree_distribution(g, mode = "in")
k = which(dd > 0) - 1
k = k[k > 0]
plot(k, dd[k + 1], log = "xy", pch = 19,
     xlab = "In-degree (log)", ylab = "Proportion of nodes (log)")
```

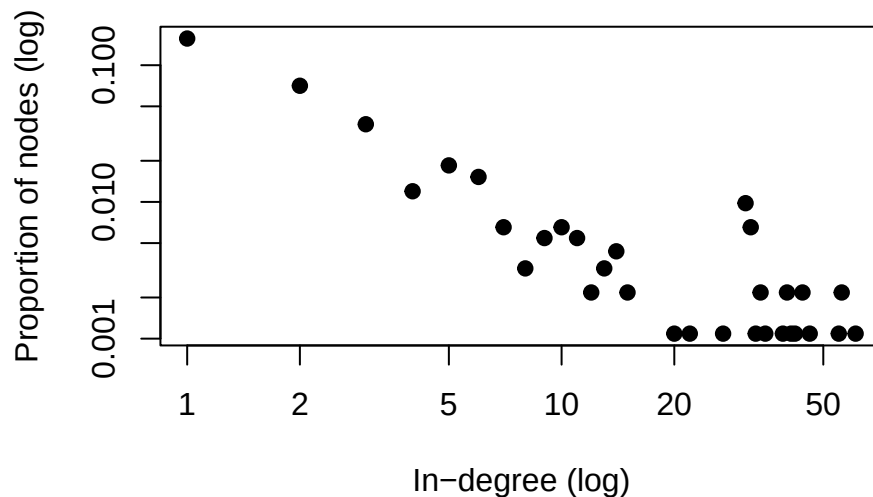


Figure 9: In-degree distribution on a log-log scale.

In-degree is heavily skewed (median 0, maximum 61) and falls roughly on a line on the log-log plot, the shape of a Barabasi-Albert graph where a few hubs get most of the links. The cluster of points around 30 is explained below.

```
hub_ids = V(g)$name[degree(g, mode = "in") >= 28 & degree(g, mode = "out") >= 28]
clique = induced_subgraph(g, hub_ids)
c(people = vcount(clique), edges = ecount(clique),
  share_of_all_edges = round(ecount(clique) / ecount(g), 2))
```

```
##           people           edges share_of_all_edges
##           31.00           930.00           0.42
```

```
table(V(clique)$gender_group)
```

```
##
## female  male
##      1    30
```

These 31 people form a complete clique (930 of 930 possible links, 42% of all links). This almost certainly comes from one navbox template rather than editors linking people one by one. 30 of the 31 are men, so we report centrality both with and without the links inside the clique.

2.5.3 Centrality

In-degree counts how many researchers' articles link to a person. PageRank also weights each link by how important the linking article is, with a random surfer who follows a link 80% of the time ($\alpha = 0.8$). PageRank scores are multiplied by the number of nodes, so 1 is average.

```
# remove only the links between clique members
g_nc = delete_edges(g, E(g)[V(g)[hub_ids] %>% V(g)[hub_ids]])

cent = data.frame(title = V(g)$title,
                  gender = V(g)$gender_group,
                  era = V(g)$era,
                  indeg = degree(g, mode = "in"),
                  pr = page_rank(g, damping = 0.8)$vector * vcount(g),
                  pr_nc = page_rank(g_nc, damping = 0.8)$vector * vcount(g))
cor(cent$indeg, cent$pr, method = "spearman")
```

```
## [1] 0.9795995
```

```
# top 10 by PageRank, with and without the clique
top = cent[order(-cent$pr), c("title", "gender", "indeg", "pr")][1:10, ]
top$pr = round(top$pr, 1)
top$no_clique_top10 = cent$title[order(-cent$pr_nc)][1:10]
rownames(top) = NULL
top
```

```
##           title gender indeg  pr
## 1      Geoffrey Hinton  male   61 12.0
## 2      Marvin Minsky    male   56 11.7
## 3 John McCarthy (computer scientist)  male   56 11.0
## 4      Demis Hassabis  male   44  8.7
## 5      Yoshua Bengio   male   55  8.5
## 6      Yann LeCun      male   46  7.8
## 7      Andrew Ng      male   40  7.3
## 8      Fei-Fei Li      female  39  7.0
## 9      Lotfi A. Zadeh   male   34  6.7
## 10     Stuart J. Russell  male   27  6.5
##           no_clique_top10
## 1      Geoffrey Hinton
## 2      Marvin Minsky
## 3 John McCarthy (computer scientist)
## 4      Stuart J. Russell
## 5      Sebastian Thrun
## 6      Lotfi A. Zadeh
## 7      Dario Amodei
## 8      Demis Hassabis
## 9      Raj Reddy
## 10     Janet L. Kolodner
```

The two measures rank people almost identically (Spearman's $\rho = 0.98$). The top 10 are the field's founders

and best-known deep learning researchers, and Fei-Fei Li is the only woman. Hinton, Minsky and McCarthy stay at the top without the clique, while Bengio, LeCun, Ng and Li drop out, so much of their centrality came from the template. Janet Kolodner enters the no-clique top 10 with only 3 incoming links because her page and Katia Sycara's link only to each other. The random surfer gets stuck in that loop, which is why we read PageRank alongside in-degree.

```
mf = cent[cent$gender != "other/non-binary", ]

# share of women among the 50 most central people (dataset average: 24%)
sapply(c(full = "pr", no_clique = "pr_nc"),
       function(v) mean(cent$gender[order(-cent[[v]])][1:50] == "female"))
```

```
##      full no_clique
##      0.14      0.16

top10  = mf$pr      >= quantile(mf$pr,      0.9)
top10_nc = mf$pr_nc >= quantile(mf$pr_nc, 0.9)
round(prop.table(table(mf$gender, top10), 1), 3)
```

```
##           top10
##           FALSE TRUE
##  female 0.963 0.037
##   male  0.880 0.120
```

```
ct = chisq.test(table(mf$gender, top10))
min(ct$expected)
```

```
## [1] 21.87132
```

```
ct
```

```
##
## Pearson's Chi-squared test with Yates' continuity correction
##
## data:  table(mf$gender, top10)
## X-squared = 11.92, df = 1, p-value = 0.0005553
```

```
chisq.test(table(mf$gender, top10_nc))$p.value
```

```
## [1] 0.0005553052
```

Centrality is too skewed to compare means, so we test whether gender is associated with being in the top 10% by PageRank.

H0: being in the top 10% by PageRank is independent of gender.

HA: the two are not independent.

Only 3.7% of women are in the top 10%, compared with 12.0% of men ($\chi^2 = 11.9$, $df = 1$, $p < 0.001$), and the result is the same without the clique ($p < 0.001$). Women are 24% of the dataset but only 14% of the 50 most central people. The gender gap is not caused by the navbox template.

```
ke = mf[mf$era != "unknown", ]
round(xtabs(pr_nc ~ era + gender, ke) / xtabs(~ era + gender, ke), 2)
```

```
##           gender
## era      female male
## 1950-69    0.96 1.12
## 1970-89    0.79 0.92
## 1990+      0.41 0.54
## pre-1950   1.07 1.45
```

```
# group sizes in the youngest era
table(ke$gender[ke$era == "1990+"])
```

```
##
```

```
## female   male
##        2    22
```

Women's mean PageRank is below men's in every era. Women reach about 74% of men's level for those born before 1950 and about 86% for 1950-89, so the gap narrows but has not closed. The 1990+ group has only 2 women, which is too few to compare.

2.5.4 Network plots

Women are orange and men are purple, which colour-blind readers can still tell apart. Node size shows PageRank on the full graph.

```
pr_full = page_rank(g, damping = 0.8)$vector
gender_col = c(female = "#E66100", male = "#5D3A9B", "other/non-binary" = "grey60")
dir.create("figures", showWarnings = FALSE)
```

The full plot shows the largest component (365 people, 97% of links), because the 474 isolated researchers would only add loose dots. We label the 8 most central women, since the top men all sit in the clique and their labels would overlap.

```
draw_full = function() {
  set.seed(3020)
  pr_l = pr_full[V(lcc)$name]
  women = V(lcc)$name[V(lcc)$gender_group == "female"]
  top_w = names(sort(pr_l[women], decreasing = TRUE))[1:8]
  plot(lcc, layout = layout_with_fr(lcc),
       vertex.color = gender_col[V(lcc)$gender_group],
       vertex.size = 1 + 7 * sqrt(pr_l / max(pr_full)),
       vertex.frame.color = NA,
       vertex.label = ifelse(V(lcc)$name %in% top_w, V(lcc)$title, NA),
       vertex.label.cex = 0.65, vertex.label.color = "black",
       vertex.label.dist = 0.8, vertex.label.font = 2,
       edge.arrow.size = 0.1, edge.color = adjustcolor("grey50", 0.3),
       main = "Largest connected component of AI researcher links (365 people)")
  legend("bottomleft", legend = c("Woman", "Man"), pch = 19, bty = "n",
        col = gender_col[c("female", "male")], pt.cex = 1.5)
}
draw_full()

png("figures/rq5_network_full.png", width = 1800, height = 1800, res = 200)
draw_full()
invisible(dev.off())
```

The dense centre is the clique, and apart from Fei-Fei Li no woman sits there. 73 of the 365 people in this component are women (20%), compared with 24% of the whole dataset.

For the subgraph we zoom in on the 40 researchers with the highest PageRank, because RQ5 asks who is central. Template links between clique members are light grey and links written by editors are dark grey. A circle layout keeps the clique from collapsing into a blob.

```
top40 = names(sort(pr_full, decreasing = TRUE))[1:40]
sub = induced_subgraph(g, top40)

# clique members first, then everyone else
ord = c(intersect(top40, hub_ids), setdiff(top40, hub_ids))
sub = permute(sub, match(V(sub)$name, ord))

# a link between two clique members comes from the navbox
from_to = ends(sub, E(sub))
is_nav = from_to[, 1] %in% hub_ids & from_to[, 2] %in% hub_ids

lay_c = layout_in_circle(sub)
```

Largest connected component of AI researcher links (365 peop

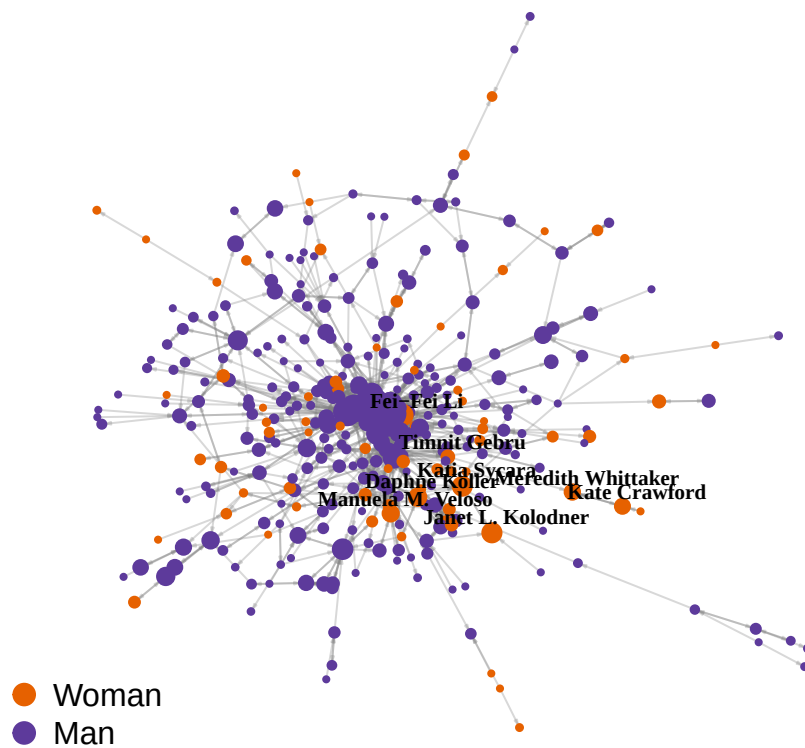


Figure 10: Largest connected component. Node size shows PageRank, colour shows gender.


```

ang = atan2(lay_c[, 2], lay_c[, 1])

draw_sub = function() {
  par(mar = c(1, 1, 3, 1))
  plot(sub, layout = lay_c, rescale = FALSE, xlim = c(-1.5, 1.5), ylim = c(-1.5, 1.5),
       vertex.color = gender_col[V(sub)$gender_group],
       vertex.size = 3 + 8 * sqrt(pr_full[V(sub)$name] / max(pr_full)),
       vertex.frame.color = NA,
       vertex.label = V(sub)$title, vertex.label.cex = 0.6,
       vertex.label.color = ifelse(V(sub)$gender_group == "female", "#B34700", "black"),
       vertex.label.font = ifelse(V(sub)$gender_group == "female", 2, 1),
       vertex.label.dist = 1.6, vertex.label.degree = -ang,
       edge.arrow.size = 0.2, edge.width = ifelse(is_nav, 0.5, 1.5),
       edge.color = ifelse(is_nav, adjustcolor("grey70", 0.35), "grey15"),
       main = "Top 40 researchers by PageRank")
  legend("bottomleft", bty = "n", cex = 0.8,
        legend = c("Woman", "Man", "Navbox (template) link", "Other link"),
        col = c(gender_col[c("female", "male")], "grey70", "grey25"),
        pch = c(19, 19, NA, NA), lty = c(NA, NA, 1, 1), lwd = 2, pt.cex = 1.5)
}
draw_sub()

png("figures/rq5_top40.png", width = 1800, height = 1800, res = 200)
draw_sub()
invisible(dev.off())

```

Only 4 of the 40 are women (Fei-Fei Li, Janet L. Kolodner, Katia Sycara, Daphne Koller). 31 of the 40 are the clique, joined by 930 template links, and only 29 editor-written links connect the other 9. Fei-Fei Li is there through the template and Daphne Koller through a few editor-written links. Kolodner and Sycara are only there because of the rank sink, so just 2 of the 40 (5%) are women with a real place in the network.

2.5.5 Article age check

An article can only collect links after it is written, and women's articles tend to be newer. We therefore check whether article age explains the gender gaps above.

```

net = data.frame(gender = V(g)$gender_group,
                 created = people$created_year[match(V(g)$name, people$qid)],
                 indeg = degree(g, mode = "in"),
                 isolated = degree(g) == 0,
                 pr = page_rank(g, damping = 0.8)$vector * vcount(g))
net = net[net$gender != "other/non-binary", ]
net$top10 = net$pr >= quantile(net$pr, 0.9)

cor(net$created, net$indeg, method = "spearman")

## [1] -0.3781707

tapply(net$created, net$gender, median)

## female male
## 2019 2014

# repeat both gender tests separately for older and newer articles
net$period = ifelse(net$created < 2015, "before 2015", "2015 onwards")
res = NULL
for (p in c("before 2015", "2015 onwards")) {
  s = net[net$period == p, ]
  ct_iso = chisq.test(table(s$gender, s$isolated))
  ct_top = chisq.test(table(s$gender, s$top10))
  res = rbind(res, data.frame(
    period = p,

```

Top 40 researchers by PageRank

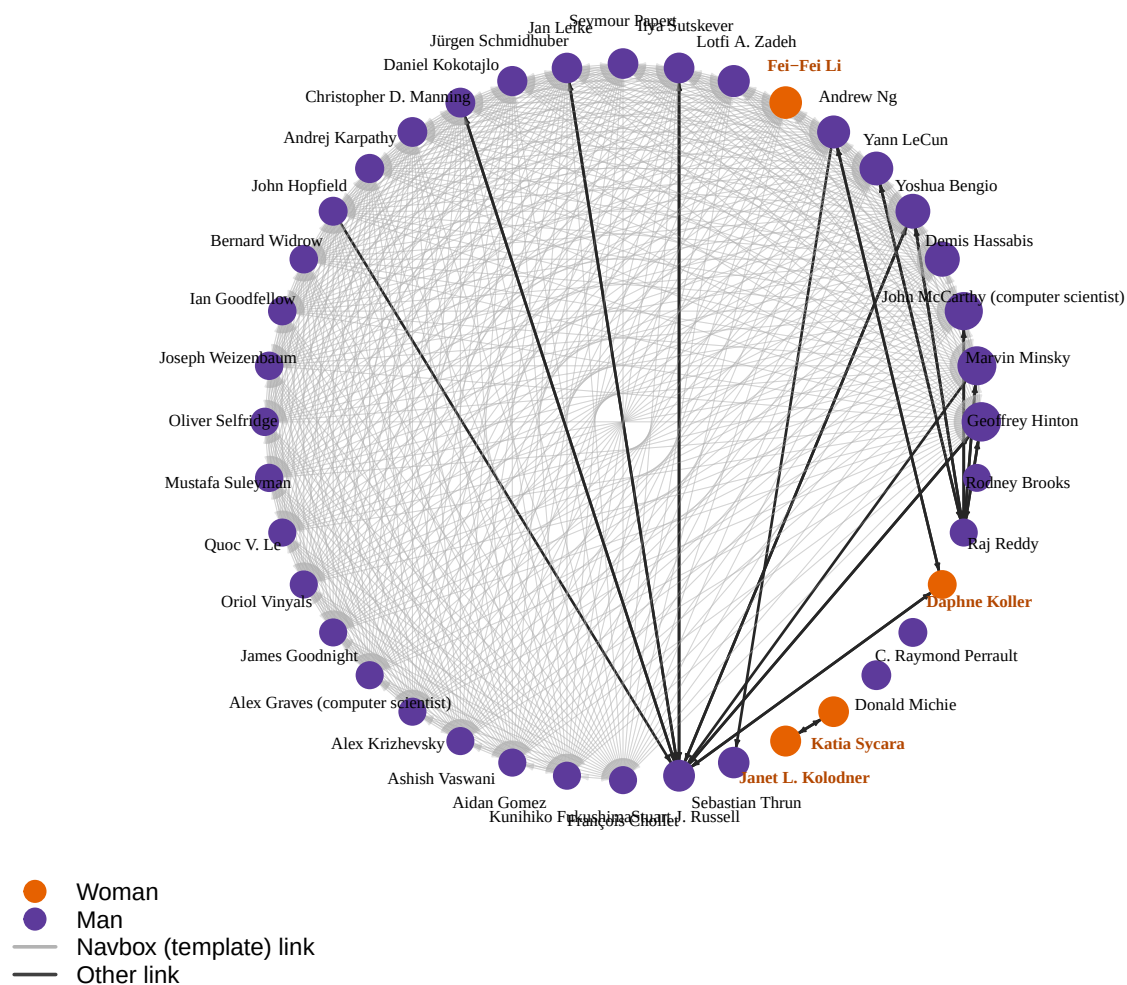


Figure 11: The 40 researchers with the highest PageRank. Light grey links come from the navbox template.

```

    women_isolated = round(mean(s$isolated[s$gender == "female"]), 2),
    men_isolated    = round(mean(s$isolated[s$gender == "male"]), 2),
    p_isolated      = round(ct_iso$p.value, 3),
    women_top10     = round(mean(s$top10[s$gender == "female"]), 3),
    men_top10       = round(mean(s$top10[s$gender == "male"]), 3),
    p_top10         = round(ct_top$p.value, 3),
    min_expected    = round(min(ct_iso$expected, ct_top$expected), 1)))
}
res

```

```

##           period women_isolated men_isolated p_isolated women_top10 men_top10
## 1   before 2015           0.33           0.38           0.583           0.140           0.155
## 2 2015 onwards           0.65           0.62           0.556           0.011           0.080
##   p_top10 min_expected
## 1   0.970           6.6
## 2   0.003           9.8

```

Newer articles get fewer links (Spearman's $\rho = -0.38$), and women's articles are newer (median 2019 vs 2014 for men). Within each period women and men are equally likely to be isolated ($p = 0.58$ and 0.56), so the isolation gap is explained by article age. The top 10% gap is not. There is no difference for articles created before 2015 (14% vs 16%, $p = 0.97$), but for articles from 2015 onwards only 1% of women reach the top 10% compared with 8% of men ($p = 0.003$). Part of this is the navbox, which includes 14 of the 26 recent men in the top 10% and no recent women, but the gap remains without it (1% vs 6%, $p = 0.03$).

3 Overall findings and limitations

3.1 Summary of each part

Data collection. We collected 919 AI researchers from Wikidata, with their article text, creation date and page views from the Wikipedia and Wikimedia APIs, and 2,204 links between their articles. Women are 24% of the dataset. After treating the placeholder birth year 2000 as unknown, 716 women and men have a known birth era.

RQ1. Gender and birth era are not independent (simulated χ^2 test, $p < 0.001$). Women's share rises from 13% (pre-1950) to 19% (1950-69) and 29% (1970-89). The 8% for 1990+ comes from only 24 people and is not a reliable reversal.

RQ2. Overall, women and men have the same recognition lag (randomisation test $p = 0.94$, Wilcoxon $p = 0.77$). Within eras, women born before 1950 ($p = 0.011$) and in 1950-69 ($p < 0.001$) received their article later in life than men, while there is no difference from 1970 onwards. The overall test hides the older gap because women are concentrated in the younger eras.

RQ3. Women and men are described with a very similar academic vocabulary (university, research, professor). The distinctive words change across eras: older women are described as advocates for women in science, women born 1950-69 by research topics and women born 1970-89 by policy and leadership roles, while the youngest men are linked to AI companies and safety.

RQ4. k-means on the tf-idf vectors gives five overlapping clusters: machine learning and statistics, language and linguistics, electrical engineering, founders and entrepreneurs, and a general AI cluster. Gender is associated with cluster (χ^2 test, $p < 0.001$). Women are almost half of language and linguistics but under a fifth of machine learning and statistics. Their share grows across eras in most clusters, but machine learning and statistics stays below 20% in every era.

RQ5. The link network is sparse (density 0.0026, 52% of researchers isolated), and 42% of its links come from one navbox clique of 31 people, 30 of them men. Only 3.7% of women are in the top 10% by PageRank, compared with 12.0% of men ($p < 0.001$), with or without the clique. Only 2 of the 40 most central researchers are women with a real position in the network. Women's higher isolation is explained by their articles being newer, but the centrality gap remains for articles created from 2015 onwards.

3.2 Limitations

Data and sample (all RQs).

- We only have researchers who are tagged with an AI field or occupation on Wikidata and who already have an English Wikipedia article. Anyone Wikipedia has missed cannot be in the data, and this could affect women more.
- Birth year is missing more often for women (35%) than men (18%), so women are under-counted in every analysis split by era.
- Gender comes from Wikidata and is almost all female or male. Only 2 people are recorded as something else, too few to analyse.
- The 1990+ era has only 24 people and 2 women, so no RQ can say much about the youngest generation (RQ1, RQ3, RQ5).
- The data is a single snapshot. Wikipedia changes every day, and requests that failed three times were skipped, so a rerun would give slightly different numbers.

Methods.

- Recognition lag (RQ2) measures when an editor wrote the article, not when the person became notable, and it is strongly affected by Wikipedia starting in 2001.
- The text analysis (RQ3) uses only the article intro, and its comparisons are descriptive and untested.
- The clusters (RQ4) overlap heavily (silhouette width at most 0.011), $k = 5$ was chosen for interpretability, and k-means can give slightly different clusters on another computer. Several cluster-era groups are small.
- The network (RQ5) only contains links between people in the dataset, so part of the isolation comes from this boundary. A link means an editor mentioned someone, not that they worked together, and links are unweighted. PageRank depends on $\alpha = 0.8$ and can be inflated by rank sinks such as the Kolodner and Sycara pair.

Interpretation. All results describe Wikipedia, not the field itself. Wikipedia editors are mostly men, so some of the gaps may reflect who writes and links articles rather than who does the research. The data is observational, so we can show associations but not causes.

3.3 Conclusion

Our question was whether the gender gap in AI is closing. The answer depends on which part of the gap we look at:

- **Representation is improving (RQ1).** Women’s share of AI researchers more than doubles from 13% for those born before 1950 to 29% for those born 1970-89.
- **Recognition has caught up (RQ2).** Women born before 1970 waited longer than men for a Wikipedia article, but there is no gap for later generations.
- **Description is mostly equal but has shifted (RQ3).** Women and men share the same academic vocabulary, and women’s distinctive words move from advocacy for women in science to research topics and leadership roles.
- **Women are unevenly spread across subfields (RQ4).** They are well represented in language and linguistics but remain under a fifth of machine learning and statistics in every era, while their share grows among founders and in general AI.
- **Centrality has not caught up (RQ5).** Women are much less likely to be among the most central researchers, and the gap is largest for articles created since 2015.

Together, the results suggest the gap is closing in numbers (RQ1) and in how quickly women are recognised (RQ2), but not in prominence (RQ5) or in the core technical subfields (RQ4). Two patterns repeat across the parts. First, overall comparisons can hide era effects: the RQ2 gap only appears within eras, and the RQ5 isolation gap disappears once article age is taken into account. Second, women’s articles are newer, so they have had less time to gain links, and the most recent generation is still too small to judge. Repeating this analysis in a few years would show whether women’s growing share turns into central positions.